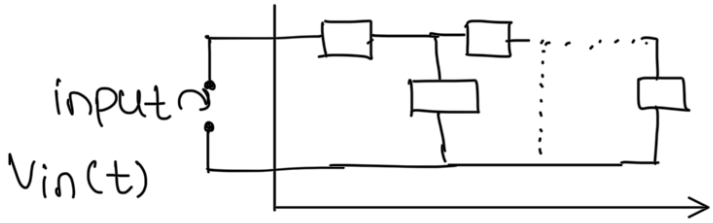
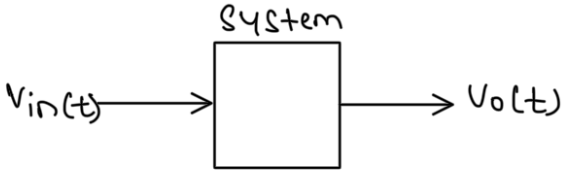
Transfer functions:



Linear Circuit

L/C/R ; linear CS

No independent source



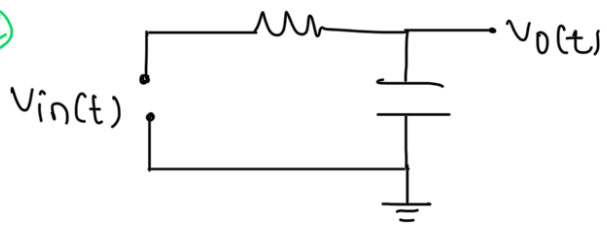
This is LTI if initially '0'

$h(t)$ ⟵ impulse response

$V_o(t) = V_{in}(t) * h(t)$

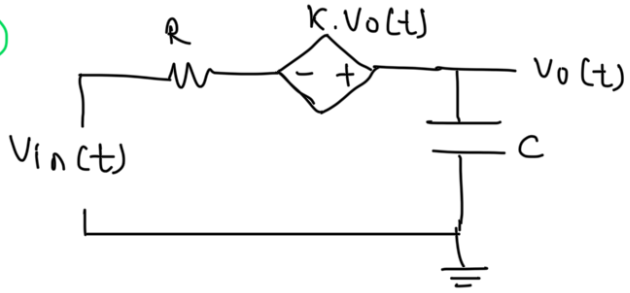
$V_o(S) = V_{in}(S) \times H(S)$ → transfer function

(I)



$$H(S) = \frac{1/SC}{R + 1/SC} = \frac{1}{1+SRC}$$

(IV)



$$i(S)\cdot R - KV_o(S) + V_o(S) = V_{in}(S)$$

$$\Rightarrow I(S) R - KI(S)\cdot\frac{1}{SC} + \frac{I(S)}{SC} = V_{in}(S)$$

$$\Rightarrow I(S) = V_{in}(S) \Big/ \left[R + \frac{1-K}{SC}\right]$$

$$V_o(S) = \frac{V_{in}(S)}{SCR+1-K} \Rightarrow H(S) = \frac{1}{SCR+1-K}$$

BIBO Statistics

LTI is BIBO if for any i/p

- $x(t)$ with "$|x(t)| < \infty \ \forall\ t$" ⇒ o/p is "$|y(t)| < \infty \ \forall\ t$" ✗
- for some B, "$|f(t)| < B \ \forall\ t$" ✓

  or $\sup |f(t)| < \infty$ or $\int |x(t)| < \infty$ ✓

∞ → infinity norm

Stability

$$\int_{-\infty}^{\infty} |h(t)|\,dt < \infty \quad \text{(if and only if)}$$

or equivalently

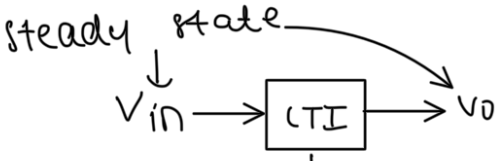
All poles of $H(s)$ are on left half of complex plane (strictly)

Poles → values of s_0 where $H(s_0) = \infty$

Circuit (I) always stable
Circuit (II) stable if $k<1$, unstable if $k>1$

Generally, dependent sources are transistor / OPAMP + seperate batteries which we do not account for

If you have passive circuit and a bit of "resistance" → stable without Resistance, just LC can be unstable



Gain system "ω dependent" ⟶ $H(s)\big|_{s=j\omega}$ steady state response

Ae^{at} → eigenfunction $a \in$ complex

$$Ae^{at} \longrightarrow \boxed{\text{LTI}} \longrightarrow H(a)\cdot Ae^{at}$$

Stable: $\text{Re}\{a\} \le 0$

$$A_1e^{j\omega t} + A_2e^{-j\omega t} \longrightarrow H(j\omega)A_1e^{j\omega t} + H(-j\omega)A_2e^{-j\omega t}$$

$$H(j\omega) = H^*(-j\omega) \qquad A_1[H(j\omega)e^{j\omega t}] + A_2[H(j\omega)e^{j\omega t}]^*$$

Plain RC circuit: $H(j\omega) = \dfrac{1}{j\omega RC + 1}$ $\quad \omega_c$ → cutoff frequency

$\tau = \frac{1}{\omega_c}$

$$= \frac{1}{j\cdot\omega/\omega_c + 1}$$

$$|H(j\omega)| = \frac{1}{\sqrt{(\omega/\omega_c)^2+1}} \qquad \angle H(j\omega) = -\tan^{-1}\left(\frac{\omega}{\omega_c}\right)$$

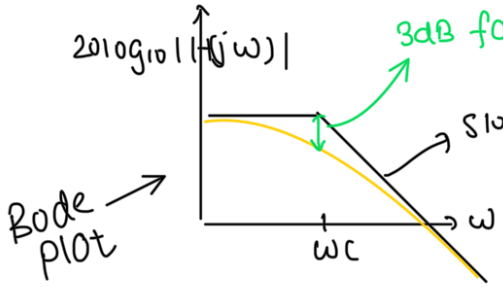
$20\log_{10}|H(j\omega)| = -20\log_{10}\left(\frac{\omega}{\omega_c}\right)$; $\omega \gg \omega_c$

$= 0$; $\omega \ll \omega_c$

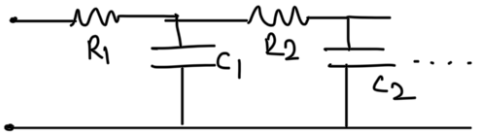
V. V imp for EE2019

$\log_{10}|H(j\omega)| \rightarrow$ unit = bel

$20\log_{10}|H(j\omega)| \rightarrow$ unit = decibel



Systems with bode plots like this



$$|H(j\omega)| = \frac{1}{SR_1C_1+1} \cdot \frac{1}{SR_2C_2+1} \cdots$$

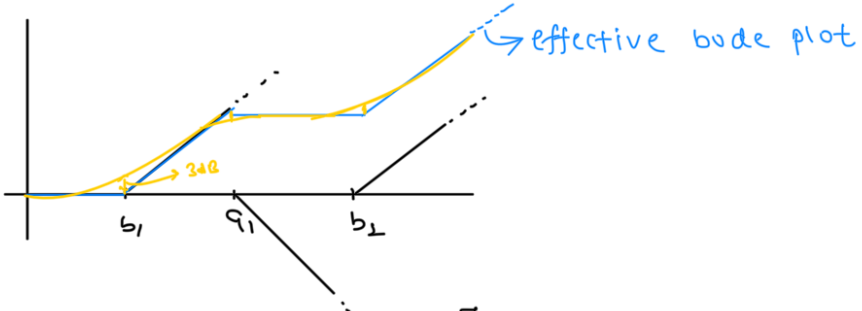
$R_1 \ll R_2 \ll \ldots$

$C_1 \gg C_2 \gg \ldots$

$$H(S) = \frac{\left(\frac{S}{b_1}+1\right)\left(\frac{S}{b_2}+1\right)\cdots}{\left(\frac{S}{a_1}+1\right)\left(\frac{S}{a_2}+1\right)\cdots}$$

assume real factors

say: $b_1 < a_1 < b_2 < a_2 \ldots$

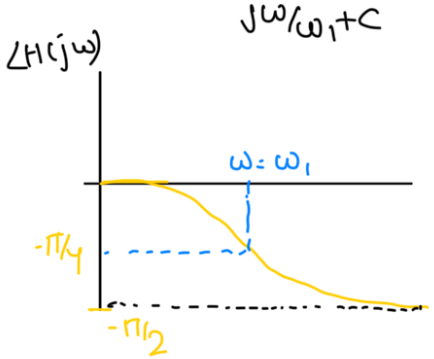


phase plot

$\frac{1}{j\omega/\omega_1+C}$

$\omega \ll \omega_1 \Rightarrow -\tan^{-1}(0)$

$\omega \gg \omega_1 \Rightarrow -\tan^{-1}(\infty) = -\pi/2$



(if $\frac{j\omega/\omega_1+C}{1}$ then 0 to $\pi/2$)